Part I — Forward Pass & Exact Chain-Rule Gradients

1. Problem Statement & Initial Conditions

Given single regression sample: $x = 1.0$, target $y = 2.0$, learning rate $\eta = 0.1$

Initial parameters at step $t = 0$: $w^{(0)} = 0.5$, $b^{(0)} = 0.0$, $v^{(0)} = 1.0$, $c^{(0)} = 0.0$

Model Architecture: $z = wx + b$, $h = \tanh(z) = \tanh(wx + b)$, $\hat{y} = vh + c$

Squared-Error Loss Function: $J(w, b, v, c) = \frac{1}{2}(\hat{y} - y)^2$

2. Step-by-Step Numerical Forward Pass at Initial Parameters

(a) Pre-activation linear combination z:

$$z = w^{(0)}x + b^{(0)} = (0.5)(1.0) + 0.0 = 0.50000000$$

(b) Hidden hyperbolic tangent activation $h = \tanh(z)$:

$$h = \tanh(0.5) = \frac{e^{0.5} - e^{-0.5}}{e^{0.5} + e^{-0.5}} = \frac{1.64872127 - 0.60653066}{1.64872127 + 0.60653066} = 0.46211716$$

(c) Linear output prediction $\hat{y}$:

$$\hat{y} = v^{(0)}h + c^{(0)} = (1.0)(0.46211716) + 0.0 = 0.46211716$$

(d) Prediction residual $e = \hat{y} - y$ and initial loss $J^{(0)}$:

$$e = \hat{y} - y = 0.46211716 - 2.0 = -1.53788284$$

Forward Pass Result: y_hat^(0) = 0.46211716, Initial Loss J^(0) = 0.5 * (-1.53788284)^2 = 1.18

3. Exact Chain-Rule Partial Derivatives (Backward Pass)

(a) Sensitivity of Loss J with respect to prediction $\hat{y}$:

$$\frac{\partial J}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}}\left[\frac{1}{2}(\hat{y} - y)^2\right] = \hat{y} - y = -1.53788284$$

(b) Gradients for output layer parameters (v, c) where $\hat{y} = vh + c$:

$$\frac{\partial J}{\partial v} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial v} = (\hat{y} - y) \cdot h = (-1.53788284)(0.46211716) = -0.71068205$$

$$\frac{\partial J}{\partial c} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial c} = (\hat{y} - y) \cdot 1 = -1.53788284$$

(c) Derivative of hidden activation $h = \tanh(z)$ with respect to z:

$$\frac{\partial h}{\partial z} = \operatorname{sech}^2(z) = 1 - \tanh^2(z) = 1 - h^2 = 1 - (0.46211716)^2 = 1 - 0.21355227 = 0.78644773$$

(d) Backpropagated hidden pre-activation sensitivity $\delta_z = \partial J / \partial z$:

$$\frac{\partial J}{\partial z} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h} \cdot \frac{\partial h}{\partial z} = (\hat{y} - y) \cdot v \cdot (1 - h^2) = (-1.53788284)(1.0)(0.78644773) = -1.20946449$$

(e) Gradients for hidden layer parameters (w, b) where $z = wx + b$:

$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial z} \cdot \frac{\partial z}{\partial w} = (-1.20946449) \cdot x = (-1.20946449)(1.0) = -1.20946449$$

$$\frac{\partial J}{\partial b} = \frac{\partial J}{\partial z} \cdot \frac{\partial z}{\partial b} = (-1.20946449) \cdot 1 = -1.20946449$$

Initial Gradients: dJ/dw = -1.20946449, dJ/db = -1.20946449, dJ/dv = -0.71068205, dJ/dc = -1.537

Part II — Simultaneous Update, Gradient Direction & FD Check

4. Simultaneous Gradient-Descent Update (Using Original Step-0 Gradients)

Rule: All four parameters update simultaneously from $\theta^{(0)}$ using $\theta^{(1)} = \theta^{(0)} - \eta \nabla_\theta J(\theta^{(0)})$ with $\eta = 0.1$:

(a) Updated hidden weight $w^{(1)}$:

$$w^{(1)} = w^{(0)} - \eta \frac{\partial J}{\partial w} = 0.5 - (0.1)(-1.20946449) = 0.5 + 0.12094645 = 0.62094645$$

(b) Updated hidden bias $b^{(1)}$:

$$b^{(1)} = b^{(0)} - \eta \frac{\partial J}{\partial b} = 0.0 - (0.1)(-1.20946449) = 0.0 + 0.12094645 = 0.12094645$$

(c) Updated output weight $v^{(1)}$:

$$v^{(1)} = v^{(0)} - \eta \frac{\partial J}{\partial v} = 1.0 - (0.1)(-0.71068205) = 1.0 + 0.07106820 = 1.07106820$$

(d) Updated output bias $c^{(1)}$:

$$c^{(1)} = c^{(0)} - \eta \frac{\partial J}{\partial c} = 0.0 - (0.1)(-1.53788284) = 0.0 + 0.15378828 = 0.15378828$$

Updated Parameters: w^(1) = 0.62094645, b^(1) = 0.12094645, v^(1) = 1.07106820, c^(1) = 0.1

5. Verification Pass After One Simultaneous Update (Step 1)

$$z^{(1)} = w^{(1)}(1.0) + b^{(1)} = 0.62094645 + 0.12094645 = 0.74189290$$

$$h^{(1)} = \tanh(0.74189290) = 0.63028688$$

$$\hat{y}^{(1)} = v^{(1)} h^{(1)} + c^{(1)} = (1.07106820)(0.63028688) + 0.15378828 = 0.82886851$$

$$J^{(1)} = \tfrac{1}{2}(0.82886851 - 2.0)^2 = \tfrac{1}{2}(-1.17113149)^2 = 0.68577449 < J^{(0)} = 1.18254182$$

6. Physical & Geometric Explanation of Gradient Direction

- *Why are all four initial partial derivatives negative?*

  At $\theta^{(0)}$, the prediction $\hat{y}^{(0)} = 0.4621$ undershoots the target $y = 2.0$, so $(\hat{y} - y) = -1.5379 < 0$.

  Since $x = 1 > 0$, $h = 0.4621 > 0$, $v = 1 > 0$, and $\operatorname{sech}^2(z) = 0.7864 > 0$, increasing ANY of (w, b, v, c) increases $\hat{y}$ toward 2.0 and therefore decreases the loss J.

- *Steepest Ascent vs. Steepest Descent:*

  The gradient vector $\nabla_\theta J(\theta^{(0)}) = [-1.2095, -1.2095, -0.7107, -1.5379]^T$ points in the direction of steepest loss INCREASE (more negative parameters $\Rightarrow$ lower $\hat{y} \Rightarrow$ larger error).

  Subtracting $\eta \nabla_\theta J$ moves in the opposite direction ($-\nabla_\theta J > 0$), increasing all four parameters simultaneously and reducing loss by 42.0% in a single step ($1.18254 \rightarrow 0.68577$).

7. Central Finite-Difference Cross-Check (C++17 Verification, epsilon = 1e-6)

Using $D_\varepsilon[J](\theta_k) = \frac{J(\theta_k + \varepsilon) - J(\theta_k - \varepsilon)}{2\varepsilon}$ with $\varepsilon = 10^{-6}$:

- $\mathrm{FD}_w = -1.2094644861$, $|\partial J/\partial w - \mathrm{FD}_w| = 1.1175 \times 10^{-10} \le 10^{-5}$ (PASS)
- $\mathrm{FD}_b = -1.2094644861$, $|\partial J/\partial b - \mathrm{FD}_b| = 1.1175 \times 10^{-10} \le 10^{-5}$ (PASS)
- $\mathrm{FD}_v = -0.7106820472$, $|\partial J/\partial v - \mathrm{FD}_v| = 4.6213 \times 10^{-11} \le 10^{-5}$ (PASS)
- $\mathrm{FD}_c = -1.5378828427$, $|\partial J/\partial c - \mathrm{FD}_c| = 2.8712 \times 10^{-11} \le 10^{-5}$ (PASS)